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Signal optimized rough silver nanoparticle for rapid SERS sensing of pesticide residues in tea

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ABSTRACT

Trace detection of toxic chemicals in foodstuffs is of great concern in recent years. Surface-enhanced Raman scattering (SERS) has drawn significant attention in the monitoring of food safety due to its high sensitivity. This study synthesized signal optimized flower-like silver nanoparticle-(AgNP) with EF at 25 °C of 1.39×10^6 to extend the SERS application for pesticide sensing in foodstuffs. The synthesized AgNP was deployed as SERS based sensing platform to detect methomyl, acetamiprid-(AC) and 2,4-dichlorophenoxyacetic acid-(2,4-D) residue levels in green tea via solid-phase extraction. A linear correlation was twigged between the SERS signal and the concentration for methomyl, AC and 2,4-D with regression coefficient of 0.9974, 0.9956 and 0.9982 and limit of detection of 5.58×10^{-4}, 1.88×10^{-4} and 4.72×10^{-3} μg/mL, respectively; the RSD value < 5% was recorded for accuracy and precision analysis suggesting that proposed method could be deployed for the monitoring of methomyl, AC and 2,4-D residue levels in green tea.

1. Introduction

Green tea is one of the leading non-alcoholic beverage in the world on account of its unique flavor and benignant medicinal activities, conserve various phenolic compounds, carbohydrate, amino acids, vitamins and minerals which aids to cure cardiovascular diseases, gastritis and various types of cancers such as liver, lung, epidermal, breast, prostate and colon; it is produced after deactivating the tea polyphenol oxidase (a heat-labile enzyme) in plucked fresh young tea leaves by introducing steam or heat (Senanayake, 2013; Zhang, Feng, Alula, & Yao, 2017; Zhu et al., 2018). It has multifarious applications as raw material in different industries like beverage, confectionery, chocolate, pharmaceuticals, cosmetics, oil, meat and so on (Senanayake, 2013). Consequently, tea should be free from toxic heavy metals, pesticides and pathogenic bacteria as well. However, the cultivation of tea is in danger due to miscellaneous diseases, weeds, and pests, causing the use of insecticide, pesticide and herbicide are regular practice to eradicate them during cultivation and also post-harvest storage.

Methomyl (S-methyl- N-(methylcarbamoyloxy)thioacetimidate), an oxime carbamate insecticide, is recurrently employed against aphids, mites, coleoptera, diptera, and hemipteran, etc. to protect agricultural crop (Rahman et al., 2017). Acetamiprid (AC) is extensively applied neonicotinoid insecticide in agricultural land owing to low mammalian and high insecticidal toxicity to control the growth of sucking and chewing pest (Ru-Yan et al., 2013). Whereas, 2,4-D, an organochlorine herbicide, is recurrently used to control the growth of broad-leaf weeds via the imitation of plant growth hormone auxin (Costa et al., 2009). They all are water soluble and typically choose water route to kill insects and weeds (Costa et al., 2017, 2009; Hassan et al., 2019a; Valencia-Quintana et al., 2016). However, AC and methomyl inhibit the catalyze activity of acetylcholinesterase (AChE) to cleavage the acetylcholine and other shorts of choline esters which acts as a neurotransmitter in the central nervous system; resulting in nerve or tissue signal transmission is interrupting which causes cell damage, hypersensitive reaction, autoimmune diseases and cancers in the body (Hassan et al., 2019a; Valencia-Quintana et al., 2016). Additionally, methomyl induces genotoxic effects such as micronuclei, sister-chromatid interchange and chromosome aberrations as well as instigates cytotoxicity through DNA damage (Guanggang et al., 2013; Valencia-Quintana et al., 2016). Meanwhile, 2,4-D treated as potential carcinogenic and mutagenic agent for the human being (Aziz et al., 2018). Methomyl has classified as a restricted insecticide by the US Environmental Protection Agency (US EPA) in 1978 (Van Scoy, Yue, Deng, & Tjeerdema, 2013). On the consideration of the toxic effect on humans, World Health Organization (WHO) has treated methomyl and 2,4-D as “highly hazardous” (Class Ib) and “moderately hazardous” (Class II)

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respectively in accordance with WHO classification scale (WHO, 2010). Maximum residual limits (MRL) have already been established for methomyl, AC and 2,4-D by different legislative frameworks. The EC (European Commission) has set MRL as 0.05, 0.05 and 0.1 μg/g, respectively for methomyl, AC and 2,4-D in tea (EC, 2019). Pesticide residues in tea might pose a potential risk for human health on account of their sub-acute and chronic toxicities and could diminish the consumption of teas or tea bags.

Up to now, different analytical techniques have been applied for the quantification of methomyl including luminescence-based terbium-3-carboxycoumarin probe (Azab et al., 2012), electro-analytical methods based on boron-doped diamond electrode (Costa et al., 2017), HPLC (high-performance liquid chromatography) coupled with fluorescence detector (Rahman et al., 2017); AC and 2,4-D such as HPLC (Geng, Chen, Wang, & Zheng, 2007; Ru-Yan et al., 2013), LC-MS/MS (Liquid chromatography/tandem mass spectrometry) (Chamkasem & Morris, 2016; Lee et al., 2017), ELISA (enzyme-linked immunosorbent assay) (Eiki, Shiro, & Yasuhiro, 2013; Wei et al., 2015). Although above all methods exhibit good selectivity and sensitivity, nevertheless, there are some shortcomings in their application, such as complicated sample preparation method, highly dependent on professional operators, time-consuming and expensive equipment. Therefore, developing a more brisk, label-free, facile, highly sensitive and cheap technique for the trace detection of methomyl, AC and 2,4-D residues levels in green tea is significant toward food safety and human health.

In recent years, vibrational spectroscopic methods, Raman spectroscopy has become an influential analytical tool for food contaminant analysis due to rapid, trace detection and high sensitivity, videlicet pesticides, heavy metals, toxins, antibiotics, pathogen, etc. (Hassan et al., 2019a; Li, Chen, Ouyang, & Zhao, 2017). The surface of noble metal nanostructure for instance silver, gold and copper exhibits potent Raman enhancement signal by virtue of its surface plasmon resonances (LSPR) (Lin, Bunker, Fernando, & Connell, 2012), which is associated with the collective oscillation of conduction electrons generated during the interaction between metal and visible or near-infrared (NIR) light (Barbosa et al., 2010). These resonances depend upon the particle size, shape, composition, dielectric properties and the neighboring medium of the metal as well (Barbosa et al., 2010). The sensitivity of nanostructure is ascertained by the degree of LSPR which is reliant on the coupling amid the electromagnetic wave and anisotropic metallic nanoarchitectures, for example, triangles, stars, cubes or flower-like nanoparticle (Barbosa et al., 2010). Consequently, the controlled-shape synthesis has gained much attention for determining SERS signatures. However, Zhang's group synthesized different SERS substrates such as graphene oxide/silver nanoparticles/silicon pyramid arrays structure (GO/Ag/PSi), AgNPs/graphene@AuNPs, ZnO nanosheets (NSs)@Ag nanoravines (NRs), Au-Ag bi-metal nanoparticles (NPs)/MoS_2 hybrid and maize-like AgNPs/PVA@Ag nanofibers for the detection of different food contaminants (Xu et al., 2018; Yu et al., 2019; Zhang et al., 2015, 2018; Zhao et al., 2019). Non-toxic, green polymers such as polyvinyl pyrrolidine act as a capping agent and for the decoration of AgNP with tuneful size, shape and optical features (Priyadarshini & Fawzy, 2017). In particular, Raman spectroscopy is an influential analytical technique owing to its high sensitivity and selectivity which can represent "fingerprint-like" information about various analyte at low concentrations, even down to a single molecule (Hassan et al., 2019a; Rycenga et al., 2011). SERS effect can be accomplished by two mechanisms, the electromagnetic (EM) and chemical (CM). The EM is the local electromagnetic field enhancement of metallic nanostructures via the incident light. Conversely, the CM is the charge transfer effect between the adsorbed particles and metallic nanostructures (Wang et al., 2020). Silver is the most significant element in plasmonics compared with Au, Cu, Al, because it can facilitate SPs in both visible and NIR regions as well as an ideal material for the electrical interconjunction ascribed to its maximum electrical and thermal conductivity among all elements. Based on its plasmonic capability, obtainable nanostructures and material price, silver is unique compared to other metals (Rycenga et al., 2011). Moreover, the AgNP can magnify light locally by 10–100 times, which basically persuade 10^6–10^8 SERS enhancement factor (Wiley, Sun, & Xia, 2007). Hassan et al. developed a simultaneous SERS based prediction method for AC and 2,4-D using gold core silver shell nanostructure (Hassan et al., 2019b). Xu et al. presented a method for rapid screening of three pesticides including 2,4-D using mesoporous silica-supported orderly-spaced gold nanoparticles as SERS substrate (Xu et al., 2020). Albeit, the reported methods were sensitive, the synthesis of SERS active substrate for those methods required multiple-step and continuous involvement, additionally, a larger number of spectral data is required to develop a chemometric model which makes them cumbersome in their real application. Therefore, simple synthesis processes crucial for practical applications.

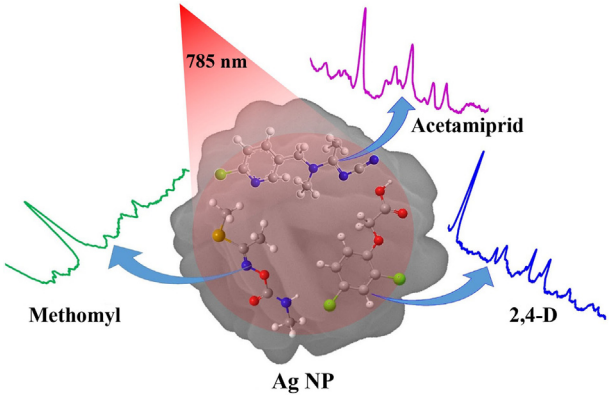
Owing to the trace level of pesticides and complexity of tea sample, a separation and pre-concentration step is desirable before analysis with high accuracy and precision. Up to now, steam distillation, soxhlet, ultrasonic, and microwave-assisted extraction, etc. are the commonly used sample extraction technique (Rodriguez, Ibarra, Galan-Vidal, Cepeda, & Miranda, 2015). Notwithstanding, techniques above are easy but they are labor-intensive, need a large volume of solvent and sample, and sometimes they are not able to remove interfering substances like natural pigments. Contrary, SPE has gained a significant attraction among the all extraction technique owing to its simplicity, accuracy, precision and fast clean-up procedure, which are commonly used in HPLC and GC–MS analysis (Gui-Jie, Wu, Wang, Hung, & Rouseff, 2019).

This paper attempted to synthesize signal optimized highly sensitive, simple and surfactant influenced flower-like AgNP through the nucleation method to analyze SERS spectral data for the rapid tracing and monitoring of methomyl, AC and 2,4-D residue levels in green tea, and SPE was introduced as an easy sample clean-up procedure to speed up the SERS detection, as indicated in Scheme 1. This study discusses (i) synthesized and characterized of flower-like AgNP at different temperature conditions as SERS substrate, (ii) fabricated label-free SERS sensor using the as-synthesized AgNP for signal optimization and application, (iii) the SERS spectrum of methomyl, 2,4-D and AC were acquired usages Raman spectrometer, (iv) SPE was applied to clean-up the tea extract for real application, (v) different statistical analyses were conducted to validate the method.

2. Materials and methods

2.1. Materials and sample

Methomyl (98.3%), AC (98.3%) and 2,4-D (98.7%) were obtained



Scheme 1. Schematic representation of the fabrication of highly roughened surface flower-like AgNP for rapid detection of methomyl, acetamiprid and 2,4-D residues levels in green tea.

from Shanghai Pesticide Research Institute Co., Ltd. (Shanghai, China). L-ascorbic acid (≥ 99%), polyvinylpyrrolidone K30 (PVP), silver nitrate ($AgNO_3$, ≥ 99.0%), crystal violet (CV), ethanol and acetonitrile (CAN) were purchased from Sinopharm Chemical Reagent Co. Ltd. (Shanghai, China). ENVI-Carb SPE (Catalog #: 57094) was obtained from Sigma-Aldrich Co. (USA). Milli-Q water was available throughout the experiment. Green (Longing) tea was collected from local Auchan supermarket and stored at 4 °C before analysis.

2.2. Instrumentation

Field Emission-Scanning Electron Microscope (FE-SEM) was used to ascertain the size and shape of flower-like AgNP (JSM-7800F, JEOL Ltd., Japan). FE-SEM sample was prepared by the below procedure, 0.1 mg/mL AgNP colloidal solution was prepared with ethanol and ultra-sonicated for 10 min after that 10 μL of this aliquot was dispersed onto a 5 × 5 mm size silicon wafer and dried at 35 °C temperature. Elemental analysis of nanoparticle was done by EDS (Energy dispersive spectroscopy) (Hitachi Ltd., Japan). In case of EDS sample preparation, we followed the above sample preparation procedure of FE-SEM, but the solution concentration was 2 mg/mL of AgNP. X-ray diffraction (XRD) pattern was acquired by Siemens D5005 instrument (Bruker AXS Ltd., Germany) to authenticate the crystal structure and the phase composition of the nanoparticles. XRD spectra were collected using dried AgNP. 785 nm incident laser light in build SPLD-RAMAN-785-Q Spectrometer was used to collect SERS spectra (Hangzhou SPL photonics Co Ltd., China). UV–Vis extinction spectrum was acquired usages UV–Visible spectroscopy (8453 UV–Vis, Agilent Technologies Inc). The size and protrusion of AgNP were measured using the FE-SEM image of AgNP via SmileView software, Ver 2.1, copyright 1998–2003 JEOL LTD.

2.3. Synthesis of flower-like AgNP

Highly roughened surface flower-like AgNP was synthesized via a modified literature procedure (Wu et al., 2017). 400 μL (1 M) $AgNO_3$ solution was mixed with 20 mL (0.167%) PVP solution under vigorous magnetic stirring, then 2 mL (0.1 M) ascorbic acid was introduced quickly to the above mixture solution and stirred for 15 min more until the solution became dark grey indicating the completion of the fabrication of flower-like AgNP. The whole reaction was accomplished at temperature of 25 °C. The prepared flower-like AgNP was separated from the reaction mixture by centrifugation at 6000 rpm for 15 min then washed with ethanol and Milli-Q water three times, respectively. The final product of AgNP was dried at 35 °C for 24 h and kept at room temperature for the next use.

2.4. Sample pretreatment

The sample pretreatment was carried out by a modified literature technique (Ru-Yan et al., 2013). In order to evaluate the method, collected green tea sample was ground and mixed thoroughly. Accurately 0.2 g ground tea sample was weighted into a screw cap test tube then spiked with different concentrations of methomyl, acetamiprid and 2,4-dichlorophenoxyacetic acid. After that, added 1.0 mL water to make it smooth paste. Then added 4.0 mL ACN and shook the mixture solution until homogeneous. The extract was obtained by centrifugation at 6000 rpm for 10 min. Again added 2.0 mL ACN into the above test tube after collecting the extract and mixed homogeneously. The second extract was collected by the above procedure and mixed with the first extract. In addition, the ENVI-carb SPE cartridge was used to remove the pigment from the extract which was activated as reported by the procedure of manufacturer before use. Afterwards, the SPE extracted sample was dried by a rotary evaporator at 40 °C under the nitrogen gas flow. Subsequently, the residue was dissolved in 200 μL of 5.0% (v/v) ethanol for SERS measurement.

2.5. SERS spectral data acquisition

Methomyl, AC and 2,4-D stock solution of 1000 μg/mL were prepared from their respective standard chemicals using 5.0% ethanol. Thereafter, the stock solution was used to prepare working solutions with concentrations of 0.0001, 0.001, 0.01, 0.1, 1, 10, 100 and 1000 μg/mL. Prior to SERS spectra collection, 8 × 8 mm size of silicon wafer was first cleaned by sequential ultra-sonication in ethanol for 15 min and thoroughly rinsed with milli-Q water. This was followed by treatment with 5.0% HNO_3 (v/v) for 30 min, then washing thoroughly with water and ultra-sonication in water for 10 min again and subsequently dried at room temperature. A 5.0 μL of 1.0 mg/mL AgNP was dropped on a cleaned silicon surface then allowed it to dry for all samples. Then each concentration of methomyl, AC and 2,4-D standard solutions were dropped on dried AgNP films separately and kept for drying. The SERS spectra were acquired from the prepared methomyl, AC and 2,4-D loaded dried SERS substrate individually. For each sample, three spectrums were acquired from three separate points and the final spectrum was obtained to make them average. The scanning range 350 to 1350 cm^{-1} with acquisition time 2 s was used to collect each spectrum. 5.0% ethanol without pesticides was treated as the blank.

3. Results and discussion

3.1. Characterization of AgNP

A highly roughened surface flower-like AgNP was synthesized by the nucleation method without applying any template or seed solution. As previously discussed, PVP influenced AgNP was fabricated in aqueous phase via the reduction of $AgNO_3$ by ascorbic acid. The fabrication of AgNP accomplished by the pre-nucleation, nucleation, and progression stage. The pre-nucleation stage started with the formation of Ag° by the reduction of Ag^{+} in the existence of ascorbic acid, Ag° agglomerated rapidly and continued until the saturation occurs. Then, the agglomerated Ag° rapidly forms a crystal nucleus, as the crystal nucleus grows, the morphology of the final crystal surface directly influenced by the progression rate of the crystal. An equal progression rate leads to isotropic progression resulting in the final particle attaining a spherical structure. In addition, non-ionic surfactant PVP also influenced the progression rate of the final crystal surface owing to its distinctive structure; it has different adhesion coefficients on different crystal surfaces. When PVP adsorbed on the crystal surface, free energy on the crystal surface changes prominently and diminishes the progression rate of the crystal surface leading to an anisotropic progression of the crystal nucleus resulting in the fabrication of flower-like AgNP (Wu et al., 2017).

Fig. 2C exhibited the XRD pattern of AgNP, four diffraction peaks at $2\theta = 38.19^{\circ}$, 44.08°, 64.51° and 77.50° were well agreed with JCPDS no. 04–0783 and assignable to the (111), (200), (220) and (311) planes of Ag (Kim, Zhang, Li, Zhang, & Jokerst, 2018). This study revealed that the flower-like AgNP was highly crystallized. To optimize the size and shape of highly roughened flower-like AgNP, three different temperature conditions, such as 15, 25 and 35 °C were used in each reaction for observing the morphology of the AgNP. At 25 °C, the AgNP was spherical in shape with a high surface roughness consisting of many orderly and disorderly protrusions as well as deep caverns between these protrusions. The mean size of the AgNP was ~ 800 nm with 15 ± 5.8 nm protrusion as captured in Fig. 1B. As shown in Fig. 1A decreased roughness and dull protrusions (11.6 ± 4.8 nm) were observed at 15 °C with a size of ~900 nm. However, when the temperature was increased to 35 °C, the large protrusions (19.9 ± 6.3 nm) were observed, and the average size of the flower-like AgNP decreased to ~750 nm (Fig. 1C). To fabricate AgNP, the proper degree of PVP coating is significant to grow sharp protrusion on Ag nuclei (Zheng, Ni, Yu, Liang, & Chen, 2016) which is directly influenced

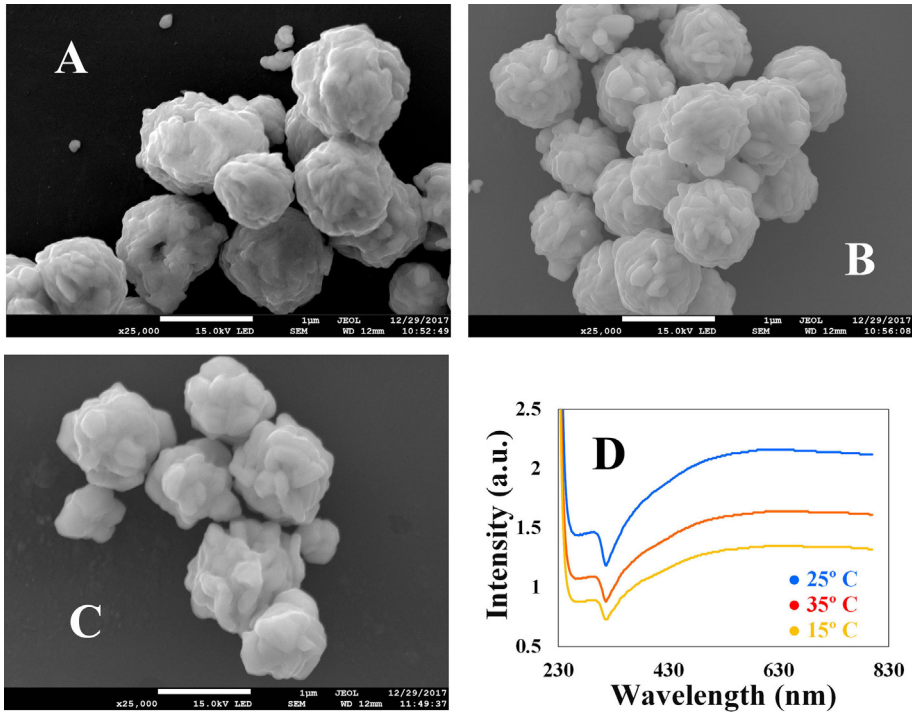


Fig. 1. The FE-SEM image of highly roughened surface AgNP synthesized at different temperature condition 15 °C (A), 25 °C (B) and 35 °C (C). UV–Vis extinction spectra of the AgNP obtained at 15, 25 and 35 °C (D).

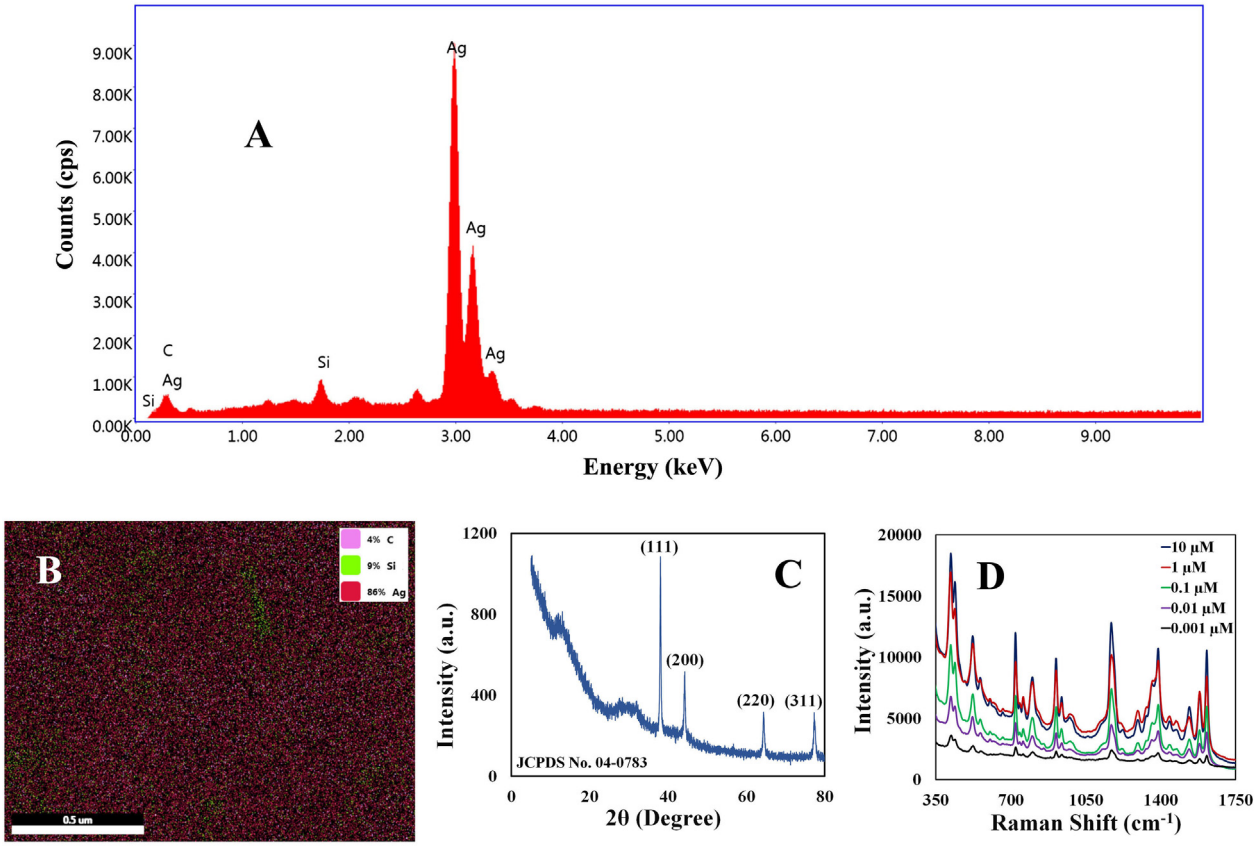


Fig. 2. EDS spectrum (A), Element overlay (B), and XRD pattern (C) of AgNP. SERS spectra of CV absorbed on AgNP (D).

by the temperature discrimination and ultimately ascertain the morphology of AgNP. Fig. 2A and B showed the histogram of EDS and element overlay respectively corroborating that silver is the principal element of AgNP. As shown in Fig. 1D, the UV–Visible extinction spectra of temperature optimized flower-like AgNP. All nanoparticle, synthesized at different temperatures such as 15, 25 and 35 °C display a band at around 294 nm owing to silver plasmon's. AgNP exhibits the strongest LSPR effect ascribed to its numerous chinks on the surface of the particle. These chinks largely enhance electromagnetic enhancement are called "hot spots effect", which ultimately provide a strong SERS effect (Taixing et al., 2013).

Crystal violet (CV), one of the extensively used SERS probes for the assessment of SERS efficiency, was selected to study the Raman response when interacting with AgNP. CV generated a number of SERS characteristic peaks over the wavelength range from 300 – 1800 cm^{-1} with different concentrations (0.001 – 10 μM) as shown in Fig. 2D. The calculated results of the enhancement factor ($EF = (I_{SERS}/C_{SERS})/(I_{RS}/C_{RS})$) for the different temperature condition synthesized AgNPs using the peak 1615 cm^{-1} of CV are summarized in Table S1. Here, C_{RS} and C_{SERS} are the lowest concertation of analyte (CV) which can generate Raman signal respectively in the absence and presence of SERS substrate (AgNP), I_{RS} and I_{SERS} are the lowest concertations of analyte (CV) corresponding Raman signal intensity respectively in the absence and presence of SERS substrate (AgNP). Based on the calculation of enhancement factor (EF), AgNP synthesized at 25 °C gave the highest EF of 1.39×10^6 whereas AgNP synthesized at 15 °C conversely had the lower EF of 4.46×10^5. In addition, AgNP synthesized at 35 °C yielded EF of 8.86×10^5 indicating that change in the temperature during AgNP synthesis could greatly impact SERS enhancement. Subsequently, 25 °C was selected as an optimum temperature to synthesis AgNP for the detection and monitoring of methomyl, AC and 2,4-D residue levels in tea.

3.2. Selection of extraction solvent

Methomyl, AC and 2,4-D are polar molecules. The mixture of water and polar organic solvent are commonly applied solvent to extract those pesticides from the food matrix. However, in order to determine the optimum extraction efficiency, two different solvent systems such as ACN and water + ACN were comparatively applied and evaluated the recovery. Compared to pure ACN, the mixture of ACN-water were exhibited optimum extraction efficiency. Additionally, various percentages of ACN (50, 60, 70, 80 and 90%) with water mixture were investigated and 80% ACN-water mixture achieved maximum extraction efficiency as shown in Table S2, which was considered as final extraction solvent for this study.

3.3. Strategy of SERS detection

As previously discussed, highly roughened surface AgNP was used as SERS substrate to detect methomyl, AC and 2,4-D residue levels in green tea. AgNP can generate strong electromagnetic enhancement by the influence of laser excitation due to its highly roughened surface area. The sulfur atom-containing imidothioate group of methomyl, the ring structure of AC and chlorine atom of 2,4-D exhibits plasmon resonance with AgNP through electromagnetic enhancement and produce strong SERS signal. A good linear relationship was realized between the SERS intensity and the concentration ranging from 1.0×10^{-3} to 1.0×10^3 μg/mL for methomyl and AC as well as 1.0×10^{-2} to 1.0×10^3 μg/mL for 2,4-D. The SERS intensity was decline with the decreasing concentration of methomyl, AC and 2,4-D. Fig. 3A shows the Raman peak of methomyl, AC and 2,4-D. Fig. 3B, C and D illustrate the typical SERS intensity for detecting different concentrations of methomyl, AC and 2,4-D using the proposed method. The SERS peak of methomyl, AC and 2,4-D were somewhat resemblance with the Raman peak. Methomyl, AC and 2,4-D gave a strong peak at 680, 628 and 392 cm^{-1} which were possibly due to c-s aliphatic bonding, N22-C21 = N17 wagging vibration mode and δ(COC) + δ(CCl), respectively. For this study, a range of concentration between 1.0×10^{-3} to 1.0×10^3 μg/mL for methomyl and AC as well as 1.0×10^{-2} to 1.0×10^3 μg/mL for 2,4-D were selected to build a calibration plot using the SERS peak 680, 628 and 392 cm^{-1}, respectively. The built calibration plot yielded a regression equation $y = 371.63\ln(x) + 2970.3$ ($R^2 = 0.9974$), $y = 384.12\ln(x) + 3496.9$ ($R^2 = 0.9956$) and $y = 396.31\ln(x) + 2320.2$ ($R^2 = 0.9982$) for methomyl, AC and 2,4-D, respectively as captured in Fig. 4A, B and C. The detection limit was found to be 5.58×10^{-4} μg/mL, 1.88×10^{-4} μg/mL and 2.87×10^{-4} μg/mL for methomyl, AC and 2,4-D, respectively following the equation: $LOD = X_a + 3S_a$, where X_a is the average concentration of 20 blank and their standard deviation is S_a. The linear range and LOD of the two pesticides quantification by this proposed method and other previously stated literature methods are compiled in Table S3 for comparative purpose. The SERS active label-free AgNP nanosensor exhibited a great improvement to detect three pesticides residue levels in green tea with comparatively low LOD except a very few methods and wide linear range than other previously reported. The fabricated proposed sensor is instinctive, inexpensive and convenient for rapid detection of methomyl, AC and 2,4-D residue levels in green tea.

3.4. Real sample analysis

Standard addition method was adopted to assess the accuracy of the proposed sensor in real sample (complex matrices). For the accuracy check, locally obtained green tea sample was spiked respectively with known concentrations (0.001, 1 and 1000 μg/mL) of methomyl and AC as well as (0.01, 1 and 1000 μg/mL) of 2,4-D solution, and extracted ten replicated spiked samples for each gradient concentration following the above-mentioned procedure. Then, the SERS spectrum was acquired using the proposed sensor and quantified the concentration of methomyl, AC and 2,4-D residue levels. The recovery result varies from over the ranges between 84.51 to 92.58%, 87.68 to 92.39% and 85.67 to 88.91% for methomyl, AC and 2,4-D, respectively as shown in Table 1, indicating the good accuracy of the proposed method.

Precision was evaluated based on the RSD (relative standard deviations) value of repeatability and reproducibility analysis. Three different concentrations of methomyl and AC (0.001, 1.0 and 1000 μg/mL) and 2,4-D (0.01, 1.0 and 1000 μg/mL) with five replicated samples were considered for each concentration in this study. Repeatability and reproducibility analysis were carried out respectively on the same day and five consecutive days. The estimated value of RSD for repeatability and reproducibility analysis were lower than 4.88% and 4.94%, respectively for three analytes as presented in Table S4. The experiment yielded an RSD value lower than 4.94%, indicating that the SERS active AgNP nano-sensor combined with SPE has a good precision to quantify methomyl, AC and 2,4-D residue levels in green tea.

4. Conclusion

In this study, an instinctive, highly sensitive and signal optimized SERS sensor based on AgNP combined with SPE has been proposed for the detection and monitoring of methomyl, AC and 2,4-D residue levels in green tea towards quality and safety assurance. The temperature optimized fabrication process of the flower-like AgNP is very simple and fast and exhibits strong LSPR effect due to its high surface roughness. AgNP was able to generate electromagnetic enhancement with methomyl, AC and 2,4-D in the presence of laser excitation and producing strong SERS signal with a wide linear range and much lower LOD than the MRL set by the EC in tea. In addition, the recovery analysis showed a satisfactory result over the range of 84.51 to 92.58%. This developed facile SPE coupled SERS method is sufficient for rapid quantitative detection of methomyl, AC and 2,4-D residue levels in

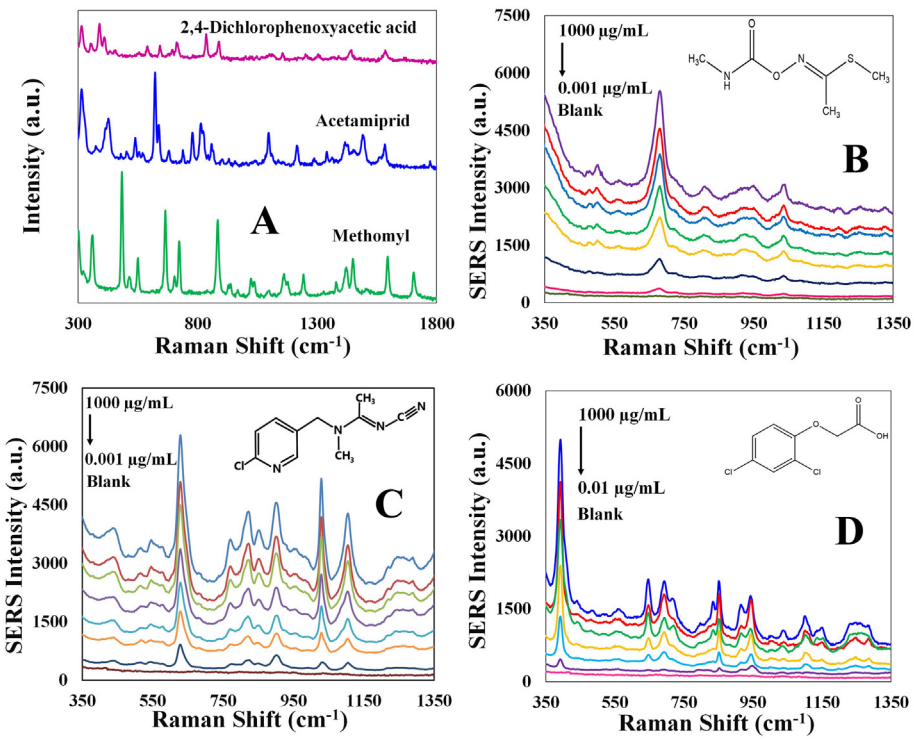


Fig. 3. Typical Raman spectrum of methomyl, acetamiprid and 2,4-D (A). SERS spectra obtained by the adsorption of AgNP with different concentrations (blank, 0.001, 0.01, 0.1, 1, 10, 100 and 1000 µg/mL) and (blank, 0.01, 0.1, 1, 10, 100 and 1000 µg/mL) of methomyl (B), acetamiprid (C) and 2,4-D (D), respectively recorded by this method.

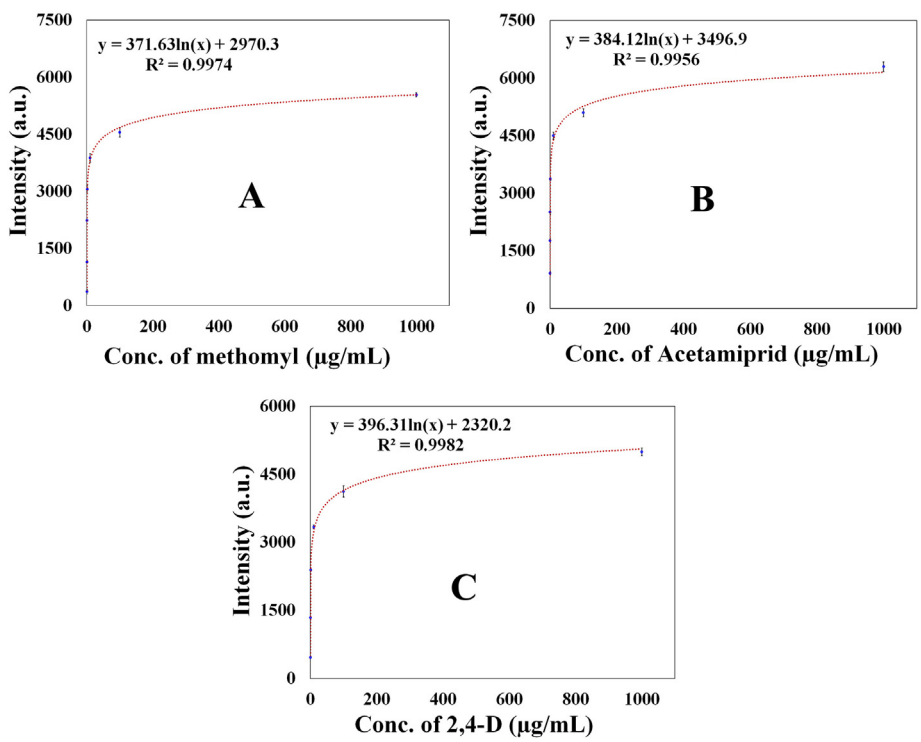


Fig. 4. Calibration plot of different concentrations vs. SERS intensity at 680, 628 and 392 cm^{-1} of methomyl (A), acetamiprid (B) and 2,4-D (C), respectively.

green tea and further study is necessary to detect more pesticides residue levels in green tea with high stability and accuracy. Moreover, this method can be applied in other food matrices by changing the sample pretreatment procedures.

CRediT authorship contribution statement

Md Mehedi Hassan: Conceptualization, Methodology, Validation, Data curation, Formal analysis, Writing - original draft. **Muhammad Zareef:** Writing - review & editing. **Tianhui Jiao:** Methodology. **Shuangshuang Liu:** Software, Validation. **Yi Xu:** Data curation, Writing - review & editing. **Annavaram Viswadevarayalu:** Software, Validation. **Huanhuan Li:** Resources. **Quansheng Chen:** Supervision.

Table 1
Determination of methomyl, AC and 2,4-D residues levels in green tea samples by this method.

Analyte	Adulterated concentration (μg/mL)	Detection value (μg/mL) (Mean[a] ± SD[b])	Recovery (%)	RSD (%)
Methomyl	0.001	0.000845 ± 0.000041	84.51	4.87
	1	0.9061 ± 0.04444	90.61	4.91
	1000	925.84 ± 14.5189	92.58	1.57
AC	0.001	0.0009239 ± 0.000045	92.39	4.89
	1	0.8768 ± 0.0415	87.68	4.73
	1000	880.38 ± 12.9755	88.04	1.47
2,4-D	0.01	0.008567 ± 0.00034	85.67	3.91
	1	0.8891 ± 0.03559	88.91	4.00
	1000	879.64 ± 14.6663	87.96	1.67

Mean[a], average value of ten tests; SD[b], standard deviation (n = 10).

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.foodchem.2020.127796.

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